

JanGyan — Learn Without Limits

Category : Cyber Safety & Digital Security

Module 4 - Banking & Payment Security

Module Overview

This handbook is your complete, beginner-to-advanced guide to **Banking & Payment Security** — covering UPI, Net Banking, QR codes, credit/debit cards, and digital wallets. India now runs on digital payments, and with that convenience comes a fast-evolving set of frauds designed to exploit small moments of trust. This handbook equips you to use every payment method confidently and safely.

By the end of this module, you will not need to search anywhere else to understand how to protect your money in India's digital payment ecosystem.

Learning Objectives

After completing this module, you will be able to:

- Explain how UPI, Net Banking, cards, and wallets each work and where their risks lie
- Understand the golden rule of QR codes and why "scan to receive" is always a scam
- Recognise how card tokenisation protects your card details
- Apply safe practices for net banking logins and sessions
- Understand digital wallet (PPI) limits and KYC rules
- Know exactly what to do — and who to contact — the moment a payment fraud happens

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Why This Topic Matters

India processed over **24,162 crore (241.62 billion) UPI transactions** worth **₹314 lakh crore** in FY2025-26 alone — a 30% year-on-year jump — making UPI the backbone of everyday Indian life, from tea stalls to temple donations. But this scale has also made digital payments the single biggest target for fraud.

> *Reality Check: UPI-related fraud cases rose from about 6.32 lakh (₹485 crore) in FY2024-25 to roughly 10.64 lakh cases (₹805 crore) in the first part of FY2025-26 — even as the underlying technology stayed secure. Almost all of this fraud succeeds by tricking the user, not by breaking UPI itself.*

Understanding exactly how each payment method is supposed to work is the best defence you have — because nearly every fraud depends on you not knowing the one rule that would have stopped it.

Introduction

Digital payments in India rely on a few core building blocks:

- **UPI (Unified Payments Interface):** An NPCI-built system that lets you transfer money instantly between bank accounts using a UPI ID or QR code.

- **Net Banking:** Accessing your bank account and moving money directly through your bank's website or app.
- **Cards (Credit/Debit):** Physical or tokenised digital cards linked to a bank account or credit line.
- **Digital Wallets (PPIs):** Prepaid Payment Instruments where you load money in advance and spend it later (e.g., mobile wallets, prepaid cards).

Each method has its own strengths, limits, and specific fraud risks — understanding these differences is the foundation of payment security.

Fundamental Concepts

Comparing India's Core Payment Methods

Method	How Money Moves	Typical Use
UPI	Directly between bank accounts, instantly	Everyday payments, P2P transfers, merchant QR payments
Net Banking	Directly from your bank account via bank's portal	Large transfers, bill payments, fixed deposits
Cards	Through a card network (Visa/Mastercard/RuPay) linked to a bank/credit line	Online shopping, POS payments, subscriptions
Wallets (PPI)	From a pre-loaded balance held by the wallet provider	Small transactions, transit, quick recharges

The Golden Rule of QR Codes

A QR code is only ever used to SEND (pay) money — never to RECEIVE it. To get paid via UPI, you don't need to scan anything; the money simply lands in your account. Any request to "scan to receive a refund/prize/payment" is a scam, without exception.

Authentication Factors in Payments

Every digital payment above a small threshold requires an **Additional Factor of Authentication (AFA)** beyond just the card or account number — such as a UPI PIN, card CVV plus OTP, or a banking password plus OTP — so that a stolen number alone cannot complete a payment.

Complete Detailed Explanation

Part A — UPI Safety

UPI moves money instantly between bank accounts using a simple UPI ID (like `name@bank`) instead of full account details.

- **UPI PIN:** A 4–6 digit PIN that authorises every payment; treat it exactly like your ATM PIN — never share it.
- **Collect Requests:** A "collect request" asks *you* to approve sending money to someone else. Fraudsters disguise payment-collect requests as "refunds" — approving one always sends money out.

- **Transaction Limits:** Generally up to ₹1 lakh per transaction/day for most personal UPI transfers, with higher limits (₹2 lakh, ₹5 lakh) permitted only for specific categories like capital markets, insurance, IPO applications, and the RBI Retail Direct Scheme.
- **Mule Accounts:** Bank accounts used to receive and quickly move fraudulent funds; RBI and NPCI have been tightening checks to detect and freeze such accounts.

Part B — Net Banking Safety

- Always type your bank's official website address directly, or use the bank's verified app — never click a banking link from an SMS or email.
- Look for "https://" and the padlock icon before entering any credentials.
- Use virtual keyboards where offered, especially on shared or public devices.
- Log out completely after every session rather than just closing the browser tab.
- Enable SMS/email alerts for all debit transactions so you notice unauthorised activity immediately.

Part C — Credit & Debit Card Safety

- **Tokenisation:** Since RBI's mandate, merchants and apps can no longer store your actual card number, expiry date, or CVV. Instead, a unique encrypted **token** — specific to that card, merchant, and device — is stored, so even a data breach at the merchant cannot expose your real card details.
- **CVV:** Never share your card's 3-digit CVV over call, SMS, or email — no bank or merchant support line legitimately needs it read aloud to you.
- **Card Controls:** Most banks let you instantly enable/disable online, international, or contactless transactions via their app — turn off what you don't actively use.
- **Card-Present vs Card-Not-Present:** Online ("card-not-present") transactions carry higher fraud risk and are why OTP/AFA is enforced for most online card payments.

Part D — Digital Wallet (PPI) Safety

Prepaid Payment Instruments (PPIs) — mobile wallets, prepaid cards, transit cards — let you load money in advance for later spending.

Wallet Type	KYC Needed	Typical Balance Cap
Small PPI (minimum KYC)	Minimal	Around ₹10,000, no cash withdrawal or fund transfer
Full-KYC PPI	Full KYC (Aadhaar/PAN-based)	Up to ₹2 lakh, supports fund transfers and interoperability
Gift/Special-Purpose PPI	Minimal	Fixed low balance, spend-only, non-transferable

The RBI has proposed further updates to these wallet rules in a 2026 draft framework (open for public comment), so always check your wallet provider's current terms for the latest applicable limits.

Important Terminologies

Term	Simple Meaning
UPI ID	A simple address (e.g., name@bank) used to send/receive UPI payments
UPI PIN	The secret PIN that authorises a UPI transaction
Collect Request	A request asking you to approve sending money — never used to receive money
Tokenisation	Replacing your real card number with an encrypted, merchant-specific token
CVV	The 3-digit security code on your card, never to be shared verbally
PPI	Prepaid Payment Instrument — a wallet or prepaid card loaded with money in advance
AFA	Additional Factor of Authentication — an extra verification step beyond the basic card/account number
Mule Account	A bank account used to receive and move fraudulent funds on a criminal's behalf
Card-Not-Present Transaction	A payment made without physically presenting the card (e.g., online)

Step-by-Step Understanding

How a Genuine UPI Payment Works (Step-by-Step)

1. You open your UPI app and enter the recipient's UPI ID or scan their QR code
2. You enter the amount and confirm the recipient's name matches who you intend to pay
3. You enter your UPI PIN to authorise the payment
4. The money moves instantly from your bank account to theirs
5. You receive an instant confirmation from your bank

What To Do Immediately After Suspected Payment Fraud

1. Call your bank's official helpline number (from your card/passbook, not a search engine) to block the card/account
2. Report the fraud on the National Cyber Crime Reporting Portal or call 1930 immediately — speed matters for fund recovery
3. Change your net banking password and UPI PIN
4. File a formal complaint with your bank in writing, referencing your cybercrime complaint number
5. Monitor your statements for any further unauthorised activity

Real-Life Indian Examples

(Widely reported public incidents, described for educational awareness.)

- **Khajuraho QR Code Swap (Madhya Pradesh):** Fraudsters pasted their own QR code stickers over the original payment codes outside multiple shops, diverting customer payments to their own accounts until a shopkeeper noticed via CCTV footage and alerted police.
- **"Scan to Receive" Marketplace Scams:** A recurring pattern on platforms like OLX, where a "buyer" asks a seller to scan a QR code to "receive" payment — the scan actually authorises an outgoing payment from the seller's account.
- **Fake Refund/Cashback QR Messages:** Messages promising a refund, cashback, or prize "if you just scan this code" have become one of the fastest-growing UPI fraud patterns reported to India's cybercrime authorities in 2025–2026.

Practical Scenarios

Scenario 1: An online "buyer" for your second-hand furniture insists you scan a QR code to receive the payment.
This is a scam. UPI never requires scanning to receive money — decline and ask for a direct UPI ID transfer instead.

Scenario 2: You get an SMS with a link to "update your net banking" before your account is "suspended."
Never click banking links from SMS. Open your bank's official app or type the website address directly instead.

Scenario 3: A caller asks you to read out your card's CVV to "verify a refund."
No refund verification ever requires your CVV. Hang up immediately — this is fraud.

Scenario 4: You've loaded ₹8,000 into a wallet app you barely use anymore.
Check the app's KYC status and consider transferring the balance out or closing dormant wallets you no longer need, reducing your exposure if that app is ever compromised.

Government Rules / Guidelines

- **RBI (Reserve Bank of India):** Regulates all digital payment systems in India, including UPI, cards, net banking, and PPIs, and issues mandatory security directions such as card tokenisation and Additional Factor Authentication.
- **NPCI (National Payments Corporation of India):** Operates and governs UPI, setting transaction limits and interoperability rules across banks and apps.
- **CERT-In:** Issues advisories on emerging payment fraud patterns, including QR code and app-based scams.

Laws & Policies

Law / Provision	What It Covers
RBI Card Tokenisation Framework	Prohibits merchants/apps from storing actual card numbers, CVV, or expiry; mandates encrypted tokens instead
RBI Authentication Mechanisms for Digital Payment Transactions Directions, 2025	Mandates two-factor authentication for all digital payments in India from 1 April 2026, with at least one dynamic factor
RBI Master Directions on Prepaid Payment Instruments	Governs KYC tiers, balance caps, and interoperability rules for digital wallets
IT Act, 2000 — Section 66C & 66D	Punish identity theft and cheating by impersonation, both commonly used in payment fraud
IT Act, 2000 — Section 43	Provides civil liability for unauthorised access or damage relating to computer systems, including banking systems

Important Statistics

Metric	Figure
UPI annual transaction volume (FY2025-26)	24,162 crore transactions
UPI annual transaction value (FY2025-26)	₹314 lakh crore
UPI YoY volume growth (2025-26)	30.0%
Average daily UPI transactions (2025-26)	About 66 crore
Banks live on UPI (March 2026)	703
UPI share of India's digital payment volume (FY2025-26)	About 85%
UPI fraud cases (FY2024-25)	About 6.32 lakh cases, ₹485 crore
UPI fraud cases (FY2025-26, till November)	About 10.64 lakh cases, ₹805 crore
Full-KYC PPI wallet balance cap	Up to ₹2 lakh
Small (minimum-KYC) PPI wallet balance cap	Around ₹10,000

(Figures are approximate, drawn from NPCI, RBI, and PIB data reported through 2025–2026.)

Best Practices

- Remember the golden rule: scanning a QR code only ever sends money, never receives it
- Access net banking only via the official app or a directly typed website address
- Never share your UPI PIN, card CVV, or net banking password/OTP with anyone
- Enable transaction alerts on every account, card, and wallet
- Turn off card features (international/online/contactless) you don't actively use
- Regularly review and close dormant wallet accounts
- Verify any "customer care" number only via the official app or website

Common Mistakes

- Scanning a QR code because someone claims it will "send you" money
- Clicking net banking links sent via SMS or WhatsApp
- Sharing CVV or OTP with a caller claiming to process a refund
- Keeping large balances in dormant, rarely used wallet apps
- Ignoring transaction alert notifications
- Searching for bank "customer care" numbers on general search engines instead of official sources

Myths vs Facts

Myth	Fact
"Scanning a QR code can help me receive money"	Scanning a QR code only ever sends money — it is never required to receive a payment
"If my card details are stored on an app, that's risky"	Since RBI's tokenisation mandate, apps store only an encrypted token, not your real card details
"UPI transactions can always be reversed easily"	Once authorised, UPI transactions are usually instant and very difficult to reverse — verify before confirming
"A wallet app is as safe as my bank account"	Wallets (PPIs) are regulated but carry different protections and limits than a full bank account
"My bank calls to verify CVV for refunds"	No bank or merchant will ever ask you to read out your CVV over a call

Do's and Don'ts

Do	Don't
Verify the payee name shown before confirming any UPI payment	Don't approve a "collect request" without checking who it's really from
Use official banking apps or directly typed URLs	Don't click net banking links from SMS or email
Keep card CVV and UPI PIN completely private	Don't share CVV, UPI PIN, or OTP with anyone, ever
Turn on transaction alerts for every account	Don't ignore unfamiliar debit notifications
Report fraud immediately via 1930/cybercrime.gov.in	Don't wait or try to resolve large fraud independently first

Safety Tips

For UPI & QR Payments
- Always check the payee name displayed before confirming any payment
- Never scan a QR code to "receive" money, a refund, or a prize
- Set a comfortable personal transaction limit within your UPI app

For Net Banking
- Bookmark your bank's official website instead of searching for it each time
- Use virtual keyboards on shared/public computers
- Always log out fully after each session

For Cards
- Disable international and contactless payments if you don't use them
- Never read your CVV aloud to anyone over a call
- Set up instant transaction alerts via SMS/app notification

For Wallets
- Complete full KYC only with wallets you actively use and trust
- Regularly clear out balances from wallets you no longer use
- Check whether a wallet is interoperable before relying on it for larger amounts

Practical Checklist

- ☐ Verified I never scan a QR code to "receive" money
- ☐ Bookmarked my bank's official net banking URL
- ☐ Enabled transaction alerts on bank account, cards, and wallets
- ☐ Reviewed and disabled unused card features (international/contactless)
- ☐ Checked KYC status and balance on all wallet apps I use

- [ ] Saved my bank's official helpline number and the 1930 cybercrime helpline
- [ ] Closed or emptied dormant wallet accounts

Quick Revision Notes

- UPI, Net Banking, Cards, and Wallets (PPIs) each work differently and carry different risks
- Golden rule: a QR code is only ever scanned to pay, never to receive money
- Card tokenisation means merchants store only an encrypted token, never your real card number/CVV
- Full-KYC wallets can hold up to ₹2 lakh; minimum-KYC wallets are capped much lower with no transfers
- RBI mandates 2FA for all digital payments in India from 1 April 2026
- Never share UPI PIN, CVV, OTP, or net banking password with anyone

Key Takeaways

1. Nearly all payment fraud succeeds by tricking you, not by breaking the technology
2. The single most protective habit: never scan a QR code to "receive" money
3. Tokenisation and 2FA are strong safety nets — but they can't protect a shared PIN, CVV, or OTP
4. Different payment methods (UPI, cards, wallets) carry different limits and protections — know which you're using
5. Speed matters after fraud — call your bank and 1930 immediately

Glossary

- **AFA (Additional Factor of Authentication)** — An extra verification step beyond the basic account/card number
- **Card-Not-Present Transaction** — A payment made without physically presenting the card
- **Collect Request** — A UPI request asking you to approve an outgoing payment
- **CVV** — The 3-digit card security code, never to be shared
- **Mule Account** — A bank account used to move fraudulent funds
- **NPCI** — National Payments Corporation of India, operator of UPI
- **PPI** — Prepaid Payment Instrument (wallet/prepaid card)
- **Tokenisation** — Replacing real card details with an encrypted, merchant-specific token
- **UPI ID** — The simple address used to send/receive UPI payments
- **UPI PIN** — The secret PIN authorising a UPI transaction

Frequently Asked Questions

1. Can scanning a QR code make me receive money?
No. A QR code is only ever used to send (pay) money — never to receive it.

2. What is the maximum UPI transaction limit?
Generally up to ₹1 lakh per transaction/day for most personal transfers, with higher limits for specific categories like capital markets or IPOs.

3. Is it safe to store my card details on shopping apps?
Yes, under RBI's tokenisation rules, apps store only an encrypted token, not your actual card number or CVV.

4. Should I ever share my CVV over a phone call?
No. No bank or merchant support line will ever legitimately ask you to read out your CVV.

5. What is a UPI collect request?
A request asking you to approve sending money to someone — it is never used to receive money, and approving a disguised one can cause a loss.

6. What should I do immediately if I'm a victim of payment fraud?
Call your bank's official helpline to block the account/card, then report on cybercrime.gov.in or call 1930 immediately.

7. Are digital wallets as regulated as banks?
Yes, PPIs (wallets) are regulated by the RBI, though with different KYC tiers and balance limits than a full bank account.

8. What is the balance limit for a Full-KYC wallet?
Up to ₹2 lakh, subject to RBI's applicable rules.

9. Can I reverse a UPI payment once it's completed?
It's usually instant and very difficult to reverse — always verify the payee before confirming.

10. What is card tokenisation?
Replacing your actual card number with an encrypted, merchant-specific token, so real card data is never stored by merchants.

11. Is net banking safe to use on public Wi-Fi?
It's best avoided; if unavoidable, use a virtual keyboard and always fully log out afterward.

12. What is a mule account?
A bank account used to receive and quickly move fraudulent funds on a criminal's behalf.

13. How can I recognise a fake QR code?
Be cautious of codes from unknown senders, codes pasted over existing ones, or any request to scan to "receive" money.

14. Do I need full KYC for a digital wallet?
Only if you want higher balance limits, fund transfers, and interoperability; minimum-KYC wallets have lower caps and fewer features.

15. What does AFA mean in digital payments?
Additional Factor of Authentication — an extra verification step (like an OTP or PIN) beyond the basic account/card number.

16. Why do online card payments need OTP but tap-and-pay sometimes doesn't?
Online ("card-not-present") transactions carry higher fraud risk, so additional authentication is enforced more strictly.

17. What happens if I click a fake net banking SMS link?
It may lead to a phishing page designed to steal your login credentials — never click such links; go to your bank's official app instead.

18. Should I turn off international transactions on my card?
Yes, if you don't travel or shop internationally, disabling this reduces your exposure to certain frauds.

19. Who regulates UPI in India?
The National Payments Corporation of India (NPCI), under the oversight of the Reserve Bank of India (RBI).

20. What's the difference between a debit card PIN and a UPI PIN?
They are different PINs for different systems, though both must be kept equally private.

21. Can closing a dormant wallet reduce my risk?
Yes, reducing the number of active financial apps and stored balances lowers your overall exposure if any one of them is compromised.

22. Where can I check the latest official UPI transaction limits?
On the NPCI website or directly within your bank's UPI app, since limits can vary by bank and category.

Suggested Infographics, Flowcharts & Illustrations

(For JanGyan design team — placement suggestions only; no images generated here)

- **"Scan to Pay, Never to Receive" visual rule card** → within "Fundamental Concepts"
- **Comparison table graphic: UPI vs Net Banking vs Cards vs Wallets** → after "Introduction"
- **Flowchart: "What To Do After Payment Fraud"** → within "Step-by-Step Understanding"
- **Card tokenisation before/after diagram** → within "Complete Detailed Explanation, Part C"
- **PPI KYC tiers comparison chart** → within "Complete Detailed Explanation, Part D"

Official References

- National Payments Corporation of India (NPCI) — UPI transaction statistics and product guidelines
- Reserve Bank of India — Card Tokenisation Framework and Card-on-File circulars
- Reserve Bank of India — Authentication Mechanisms for Digital Payment Transactions Directions, 2025
- Reserve Bank of India — Master Directions on Prepaid Payment Instruments (including the 2026 draft revision)
- Press Information Bureau (PIB), Government of India — releases on UPI growth and digital payment security
- Information Technology Act, 2000 (as amended), India Code portal

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Official Government Website Links

- Reserve Bank of India: **https://www.rbi.org.in**
- NPCI (UPI): **https://www.npci.org.in**
- CERT-In: **https://www.cert-in.org.in**
- National Cyber Crime Reporting Portal: **https://cybercrime.gov.in**
- Digital India: **https://www.digitalindia.gov.in**

Cybercrime Helpline (24×7, Toll-Free): 1930

Conclusion

UPI, net banking, cards, and wallets have made India one of the most digitally advanced payment ecosystems in the world — and the underlying technology is genuinely secure. Almost every fraud succeeds not by breaking that technology, but by convincing a person to hand over a PIN, OTP, CVV, or an unnecessary QR scan. Knowing the golden rules in this handbook puts that decision back in your hands.

This is only the next step in your Cyber Safety journey with JanGyan. Stay alert, stay informed, and remember: **you scan to pay, never to receive — and you never share your PIN, CVV, or OTP.**

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